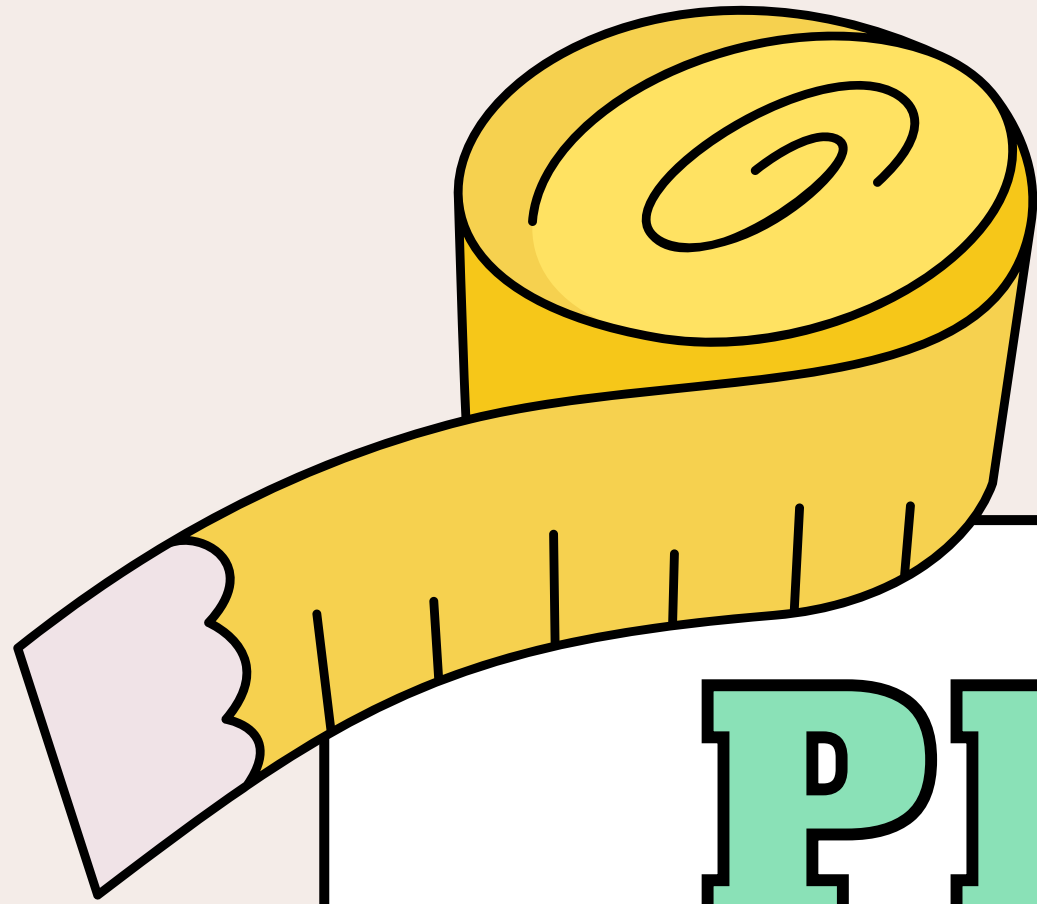




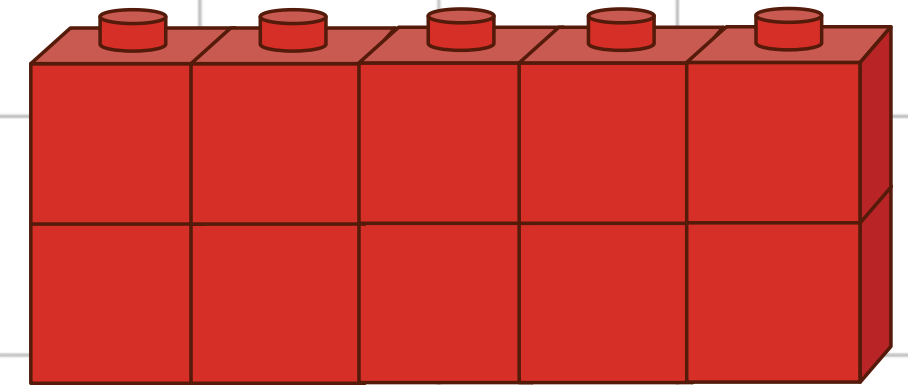
PERIMETER AND AREA

GRADE 4

What is an Irregular Shape ?

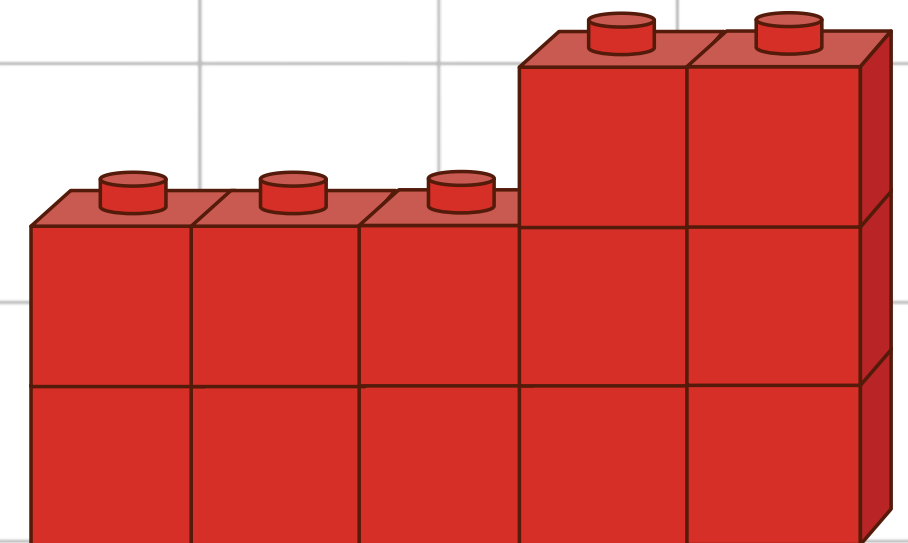
Regular Shapes: Have clear formulas (e.g., rectangle = $L \times W$).

Regular Shape



Irregular Shapes: Made up of different parts that we must break down to measure.

Irregular Shape



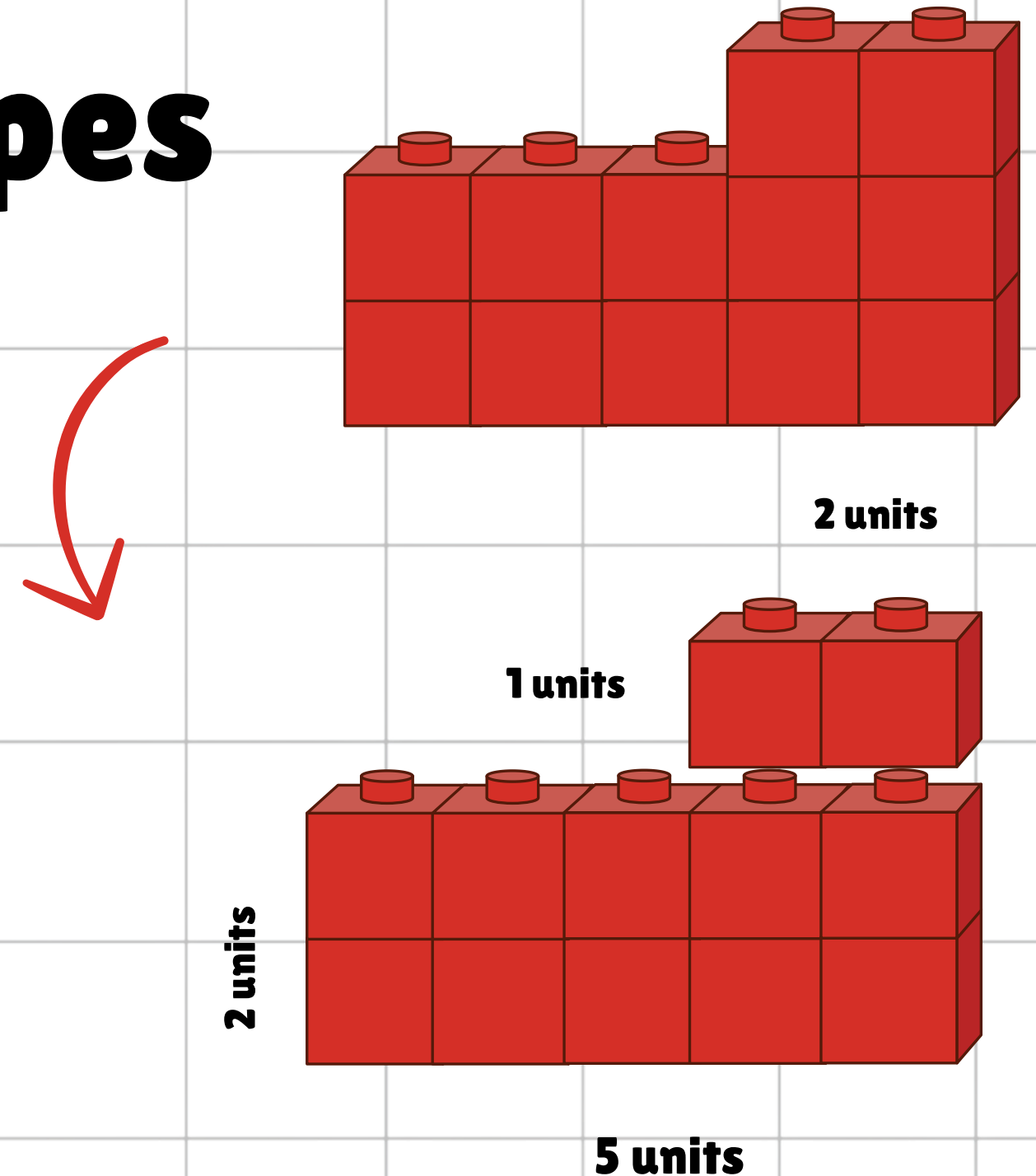
Area: Strategy 1

Break It into Smaller Shapes

Step 1: Divide the irregular shape into rectangles or squares.

Step 2: Find the area of each smaller shape using $L \times W$.

Step 3: Add up all the areas to get the total.



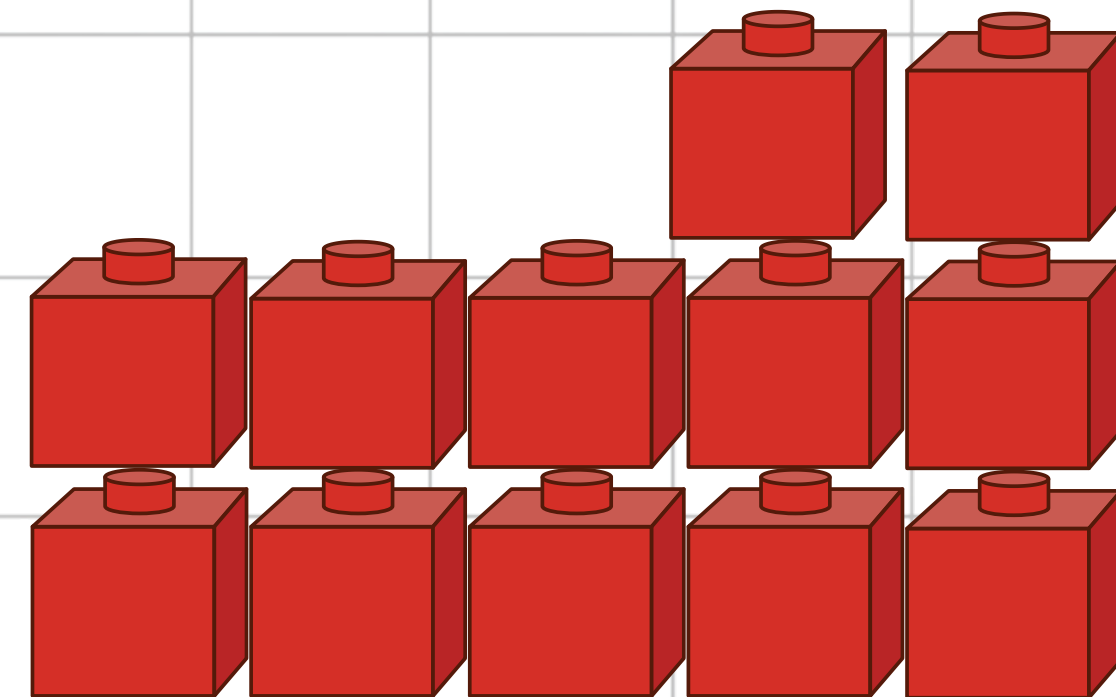
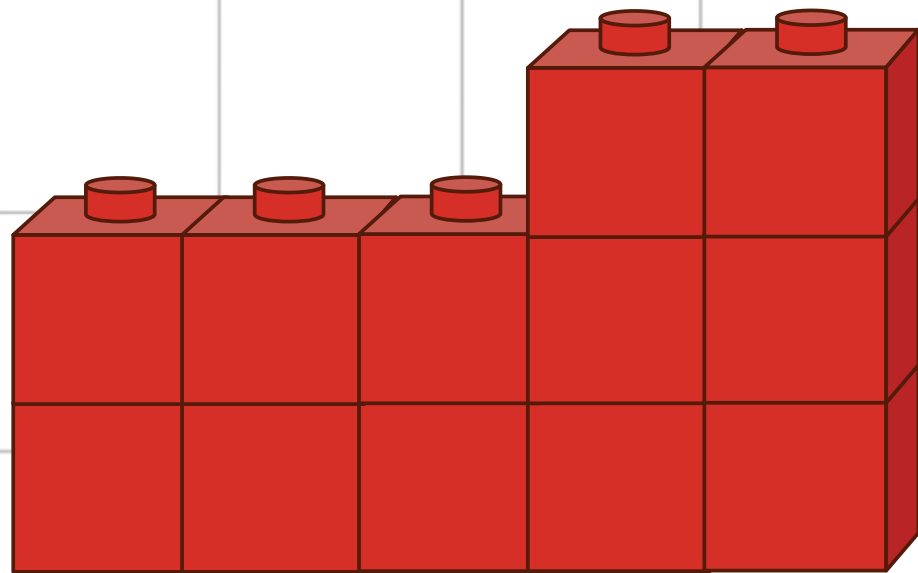
$$(2 \times 5) + (2 \times 1) = 12 \text{ square units}$$

Area: Strategy 2

Count the Squares

What if your shape is too tricky to split?

- Count full squares inside the shape.

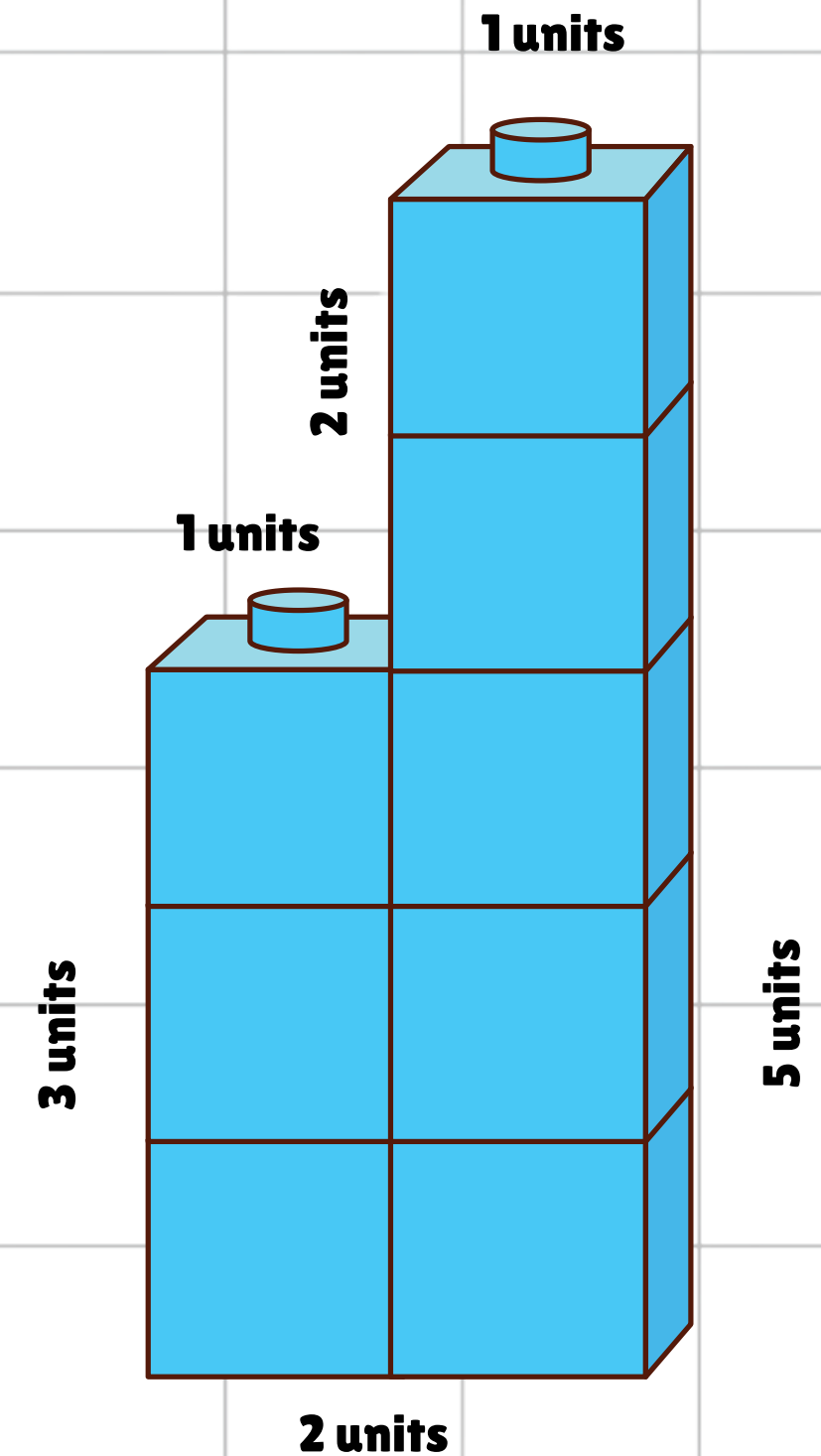


Perimeter: Strategy 2

Add Up the Sides

Step 1: Find and label the length of each side.

Step 2: Add all the sides together to get the total perimeter.



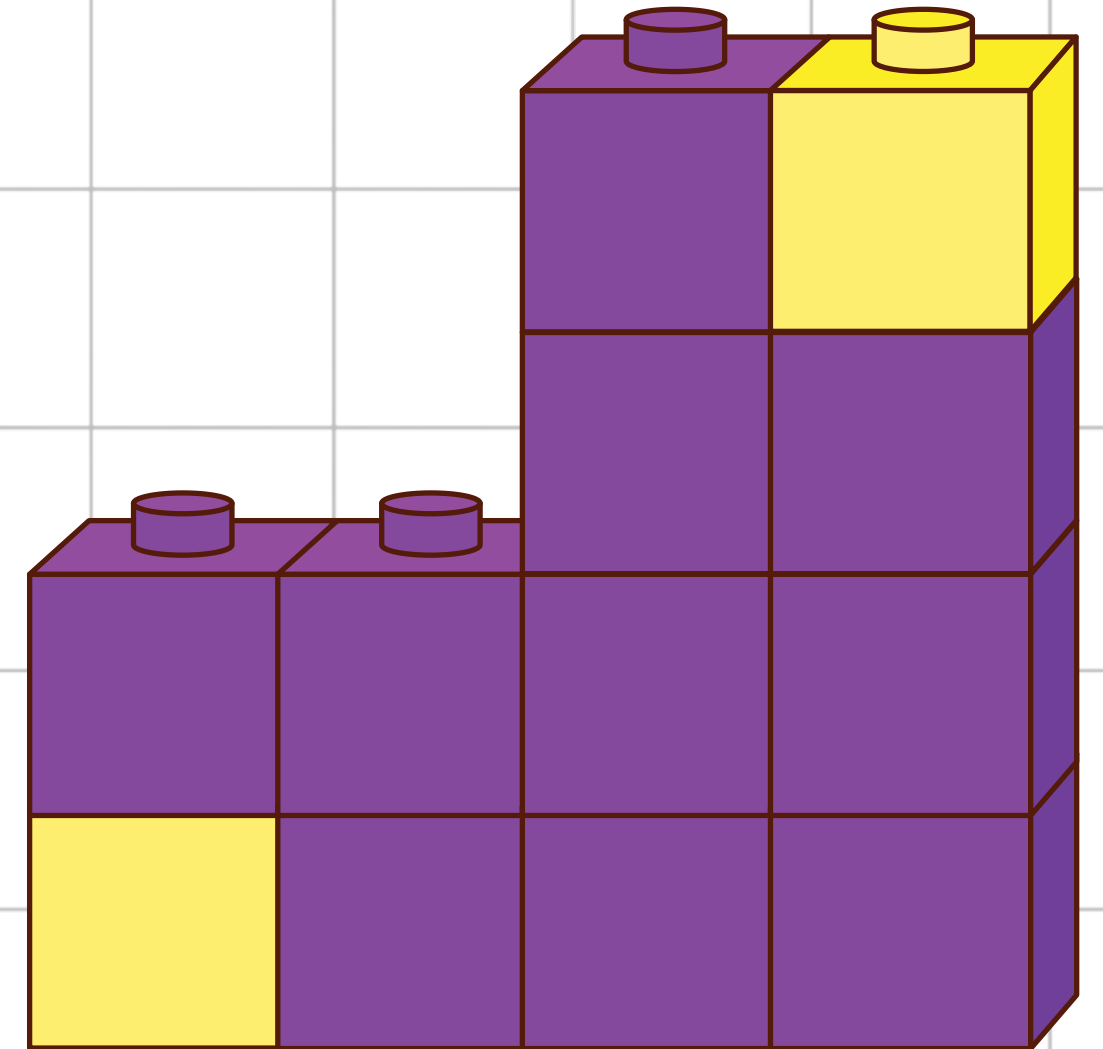
$$3 + 2 + 1 + 2 + 1 + 2 + 5 = 16 \text{ units}$$

Find the Area and Perimeter of the figure below?

Measure each shape to determine the perimeter and area. Each block represents 1 unit.

Perimeter : _____

Area: _____

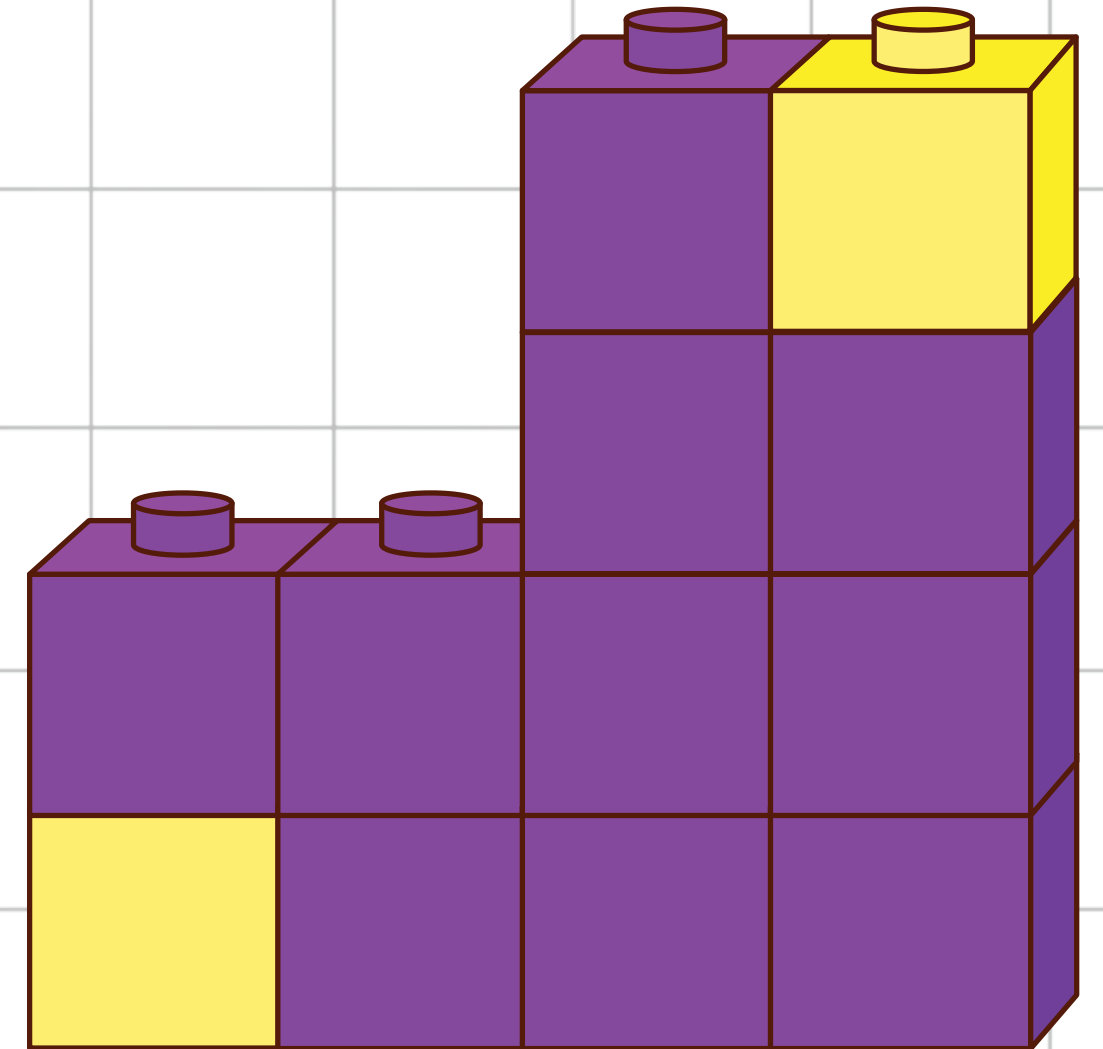


Find the Area and Perimeter of the figure below?

Measure each shape to determine the perimeter and area. Each block represents 1 unit.

Perimeter : 16 units

Area: 12 square units

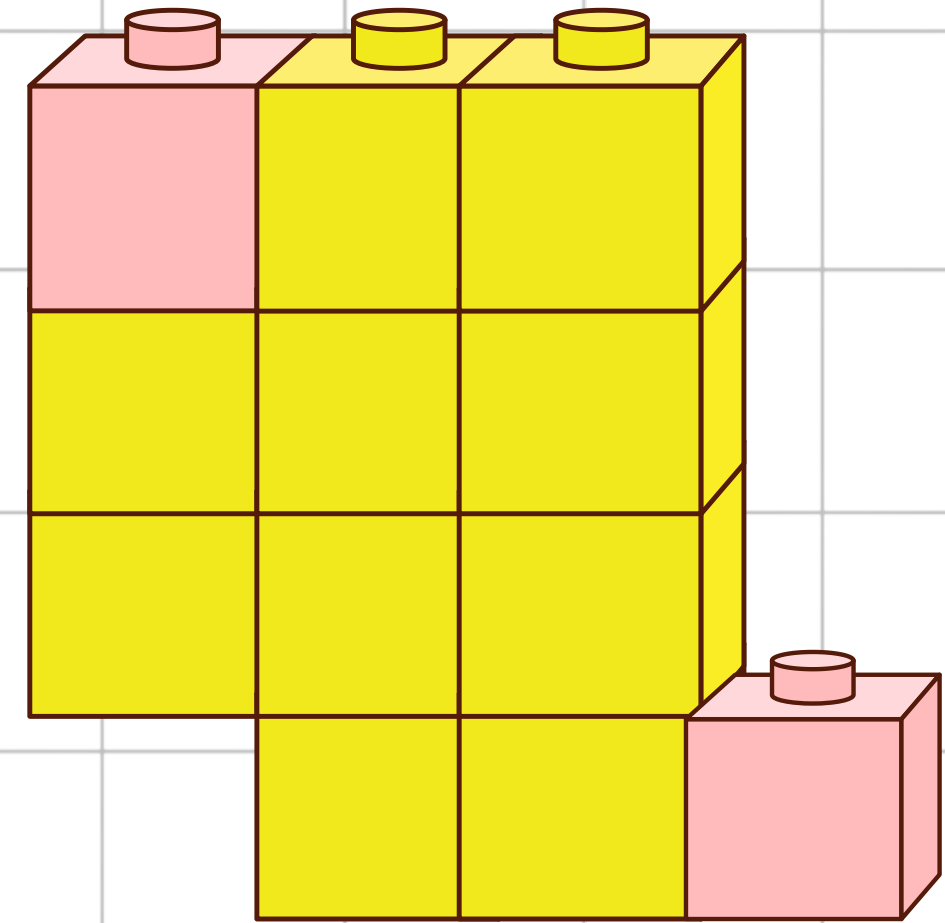


Find the Area and Perimeter of the figure below?

Measure each shape to determine the perimeter and area. Each block represents 1 unit.

Perimeter : _____

Area: _____

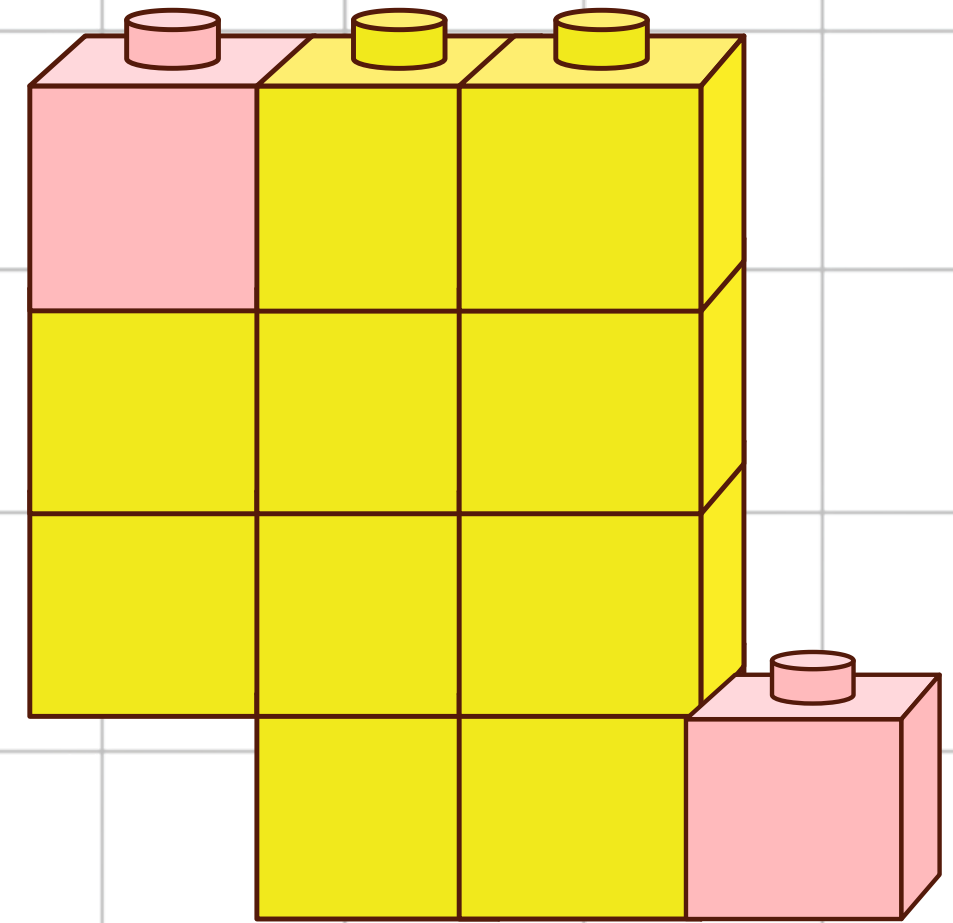


Find the Area and Perimeter of the figure below?

Measure each shape to determine the perimeter and area. Each block represents 1 unit.

Perimeter : 16 units

Area: 12 units

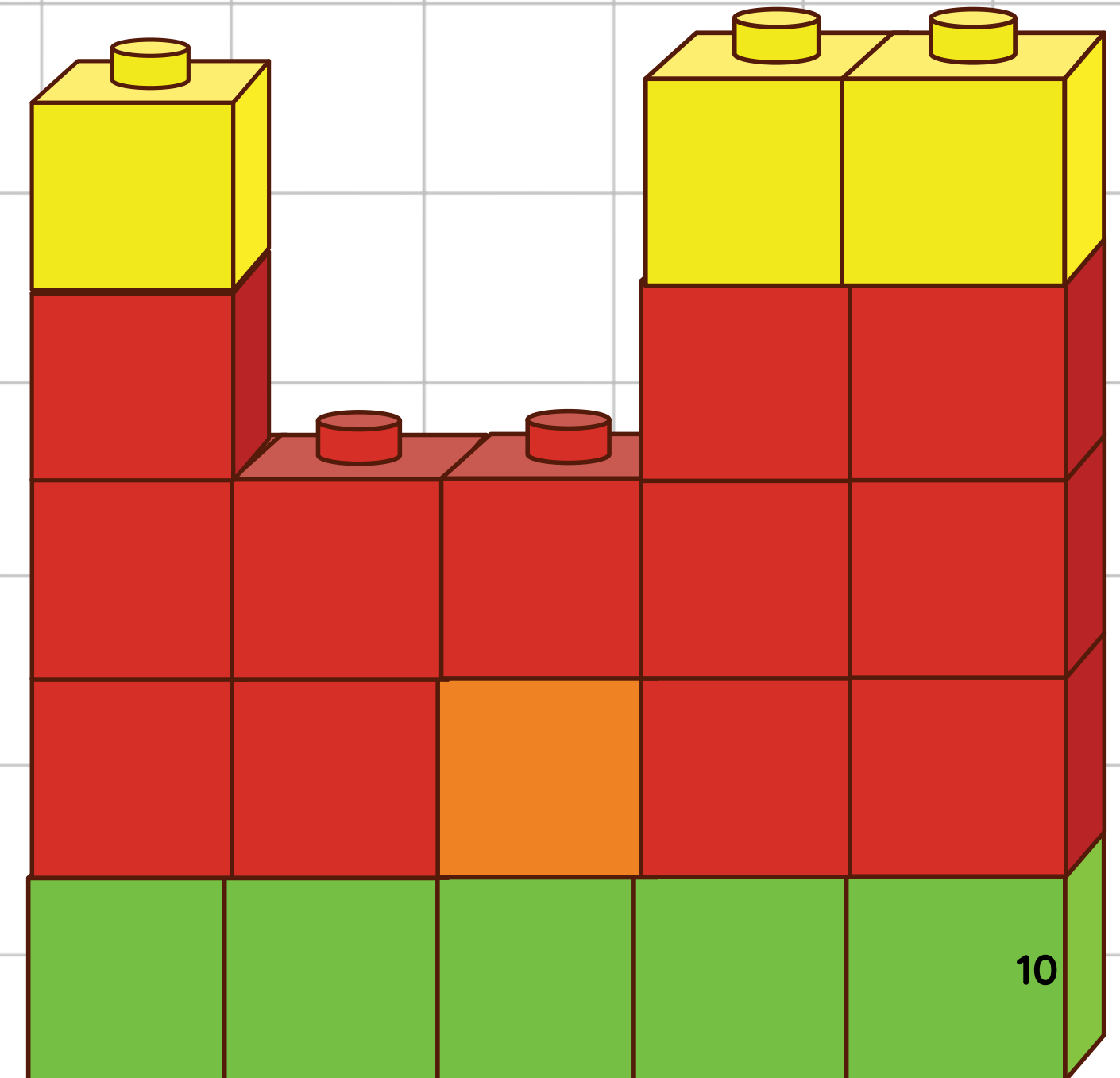


Find the Area and Perimeter of the figure below?

Measure each shape to determine the perimeter and area. Each block represents 1 unit.

Perimeter : _____

Area: _____

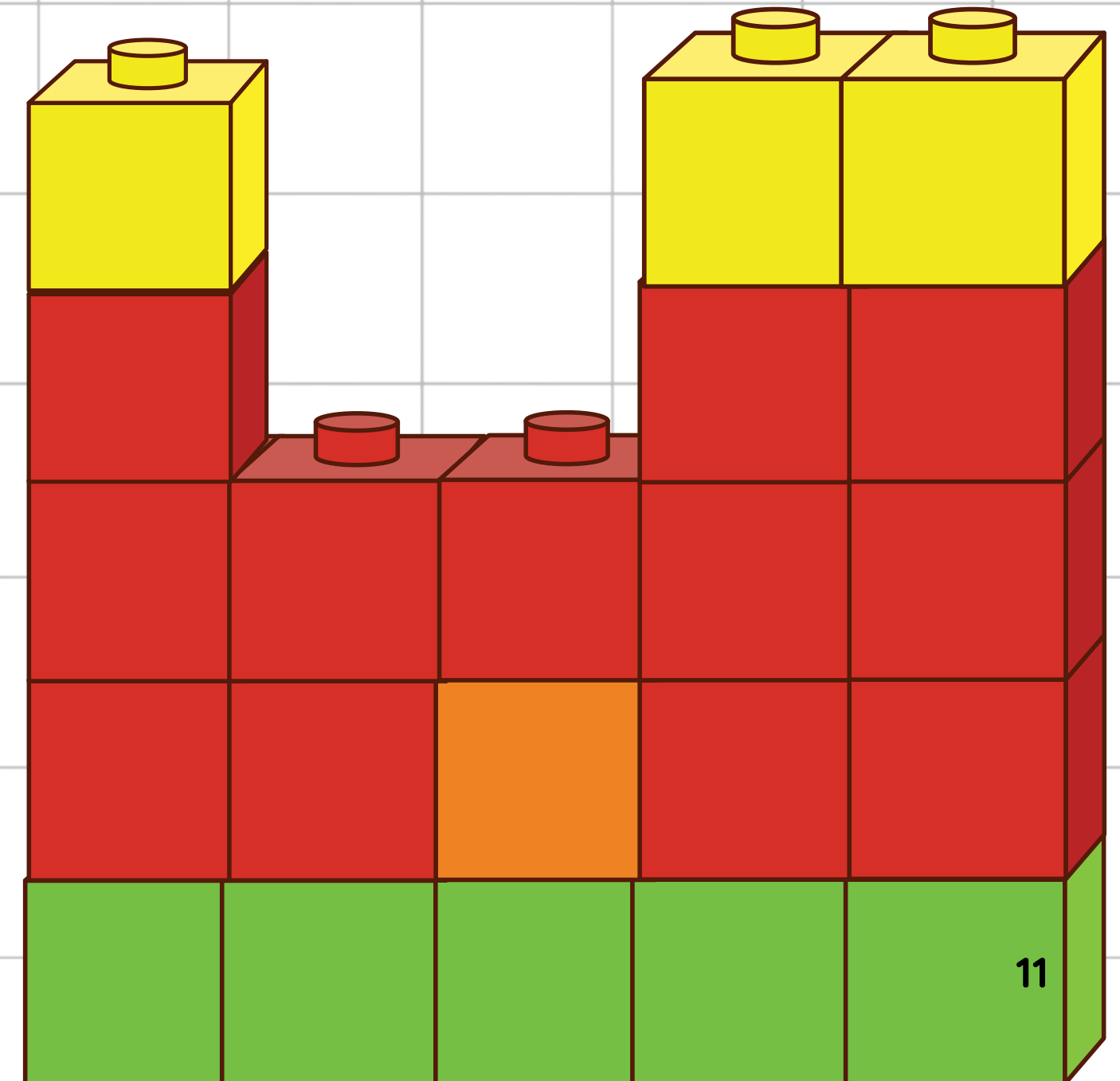


Find the Area and Perimeter of the figure below?

Measure each shape to determine the perimeter and area. Each block represents 1 unit.

Perimeter : 24 units

Area: 21 units

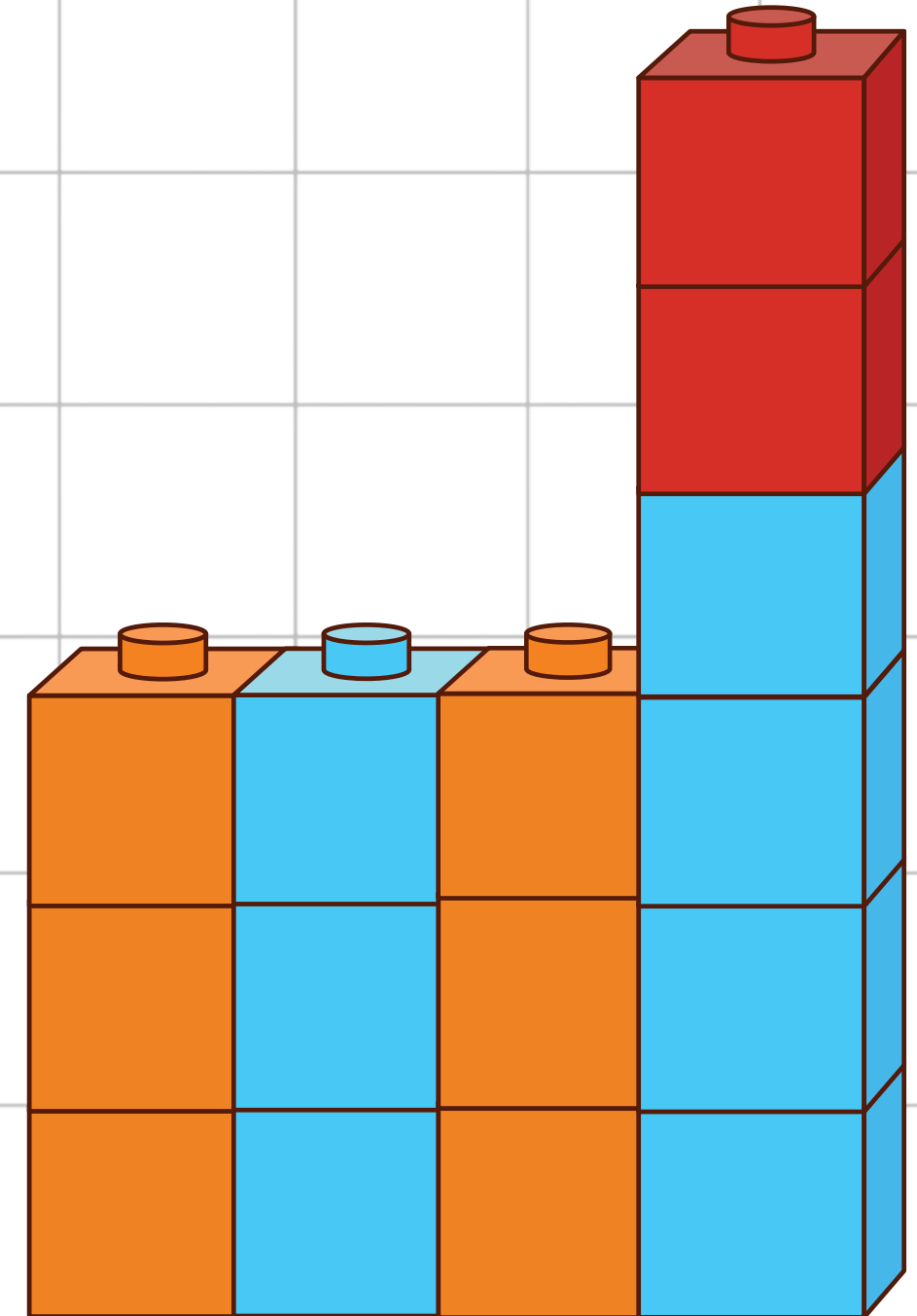


Find the Area and Perimeter of the figure below?

Measure each shape to determine the perimeter and area. Each block represents 1 unit.

Perimeter : _____

Area: _____

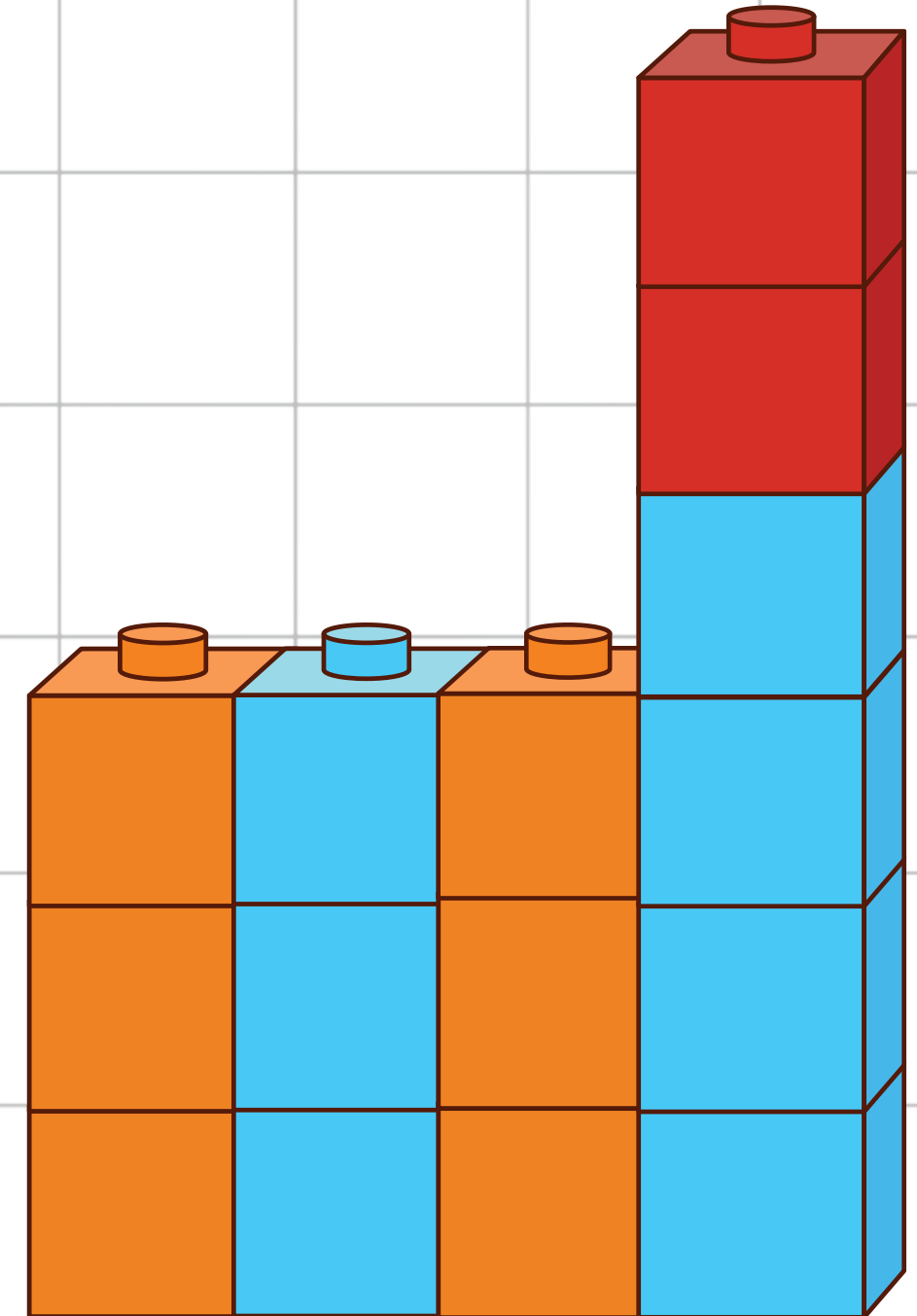


Find the Area and Perimeter of the figure below?

Measure each shape to determine the perimeter and area. Each block represents 1 unit.

Perimeter : 20 units

Area: 15 units

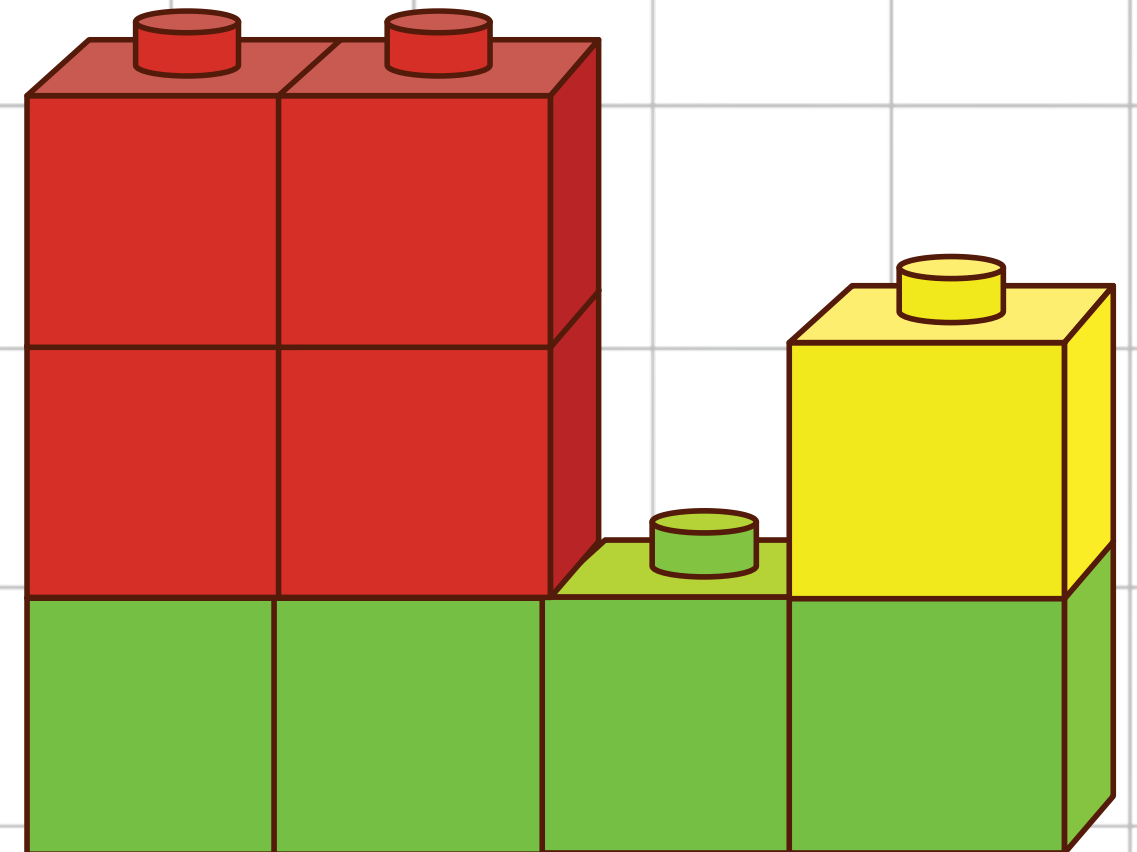


Find the Area and Perimeter of the figure below?

Measure each shape to determine the perimeter and area. Each block represents 1 unit.

Perimeter : _____

Area: _____

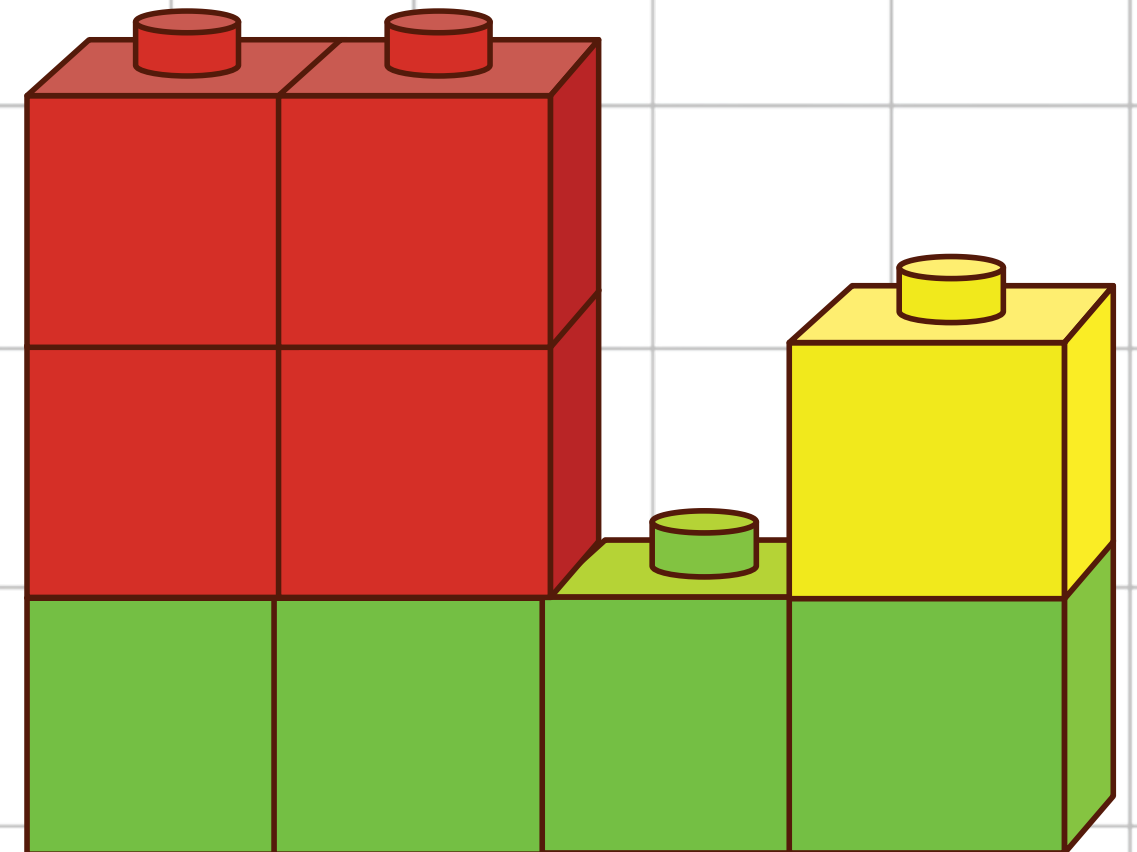


Find the Area and Perimeter of the figure below?

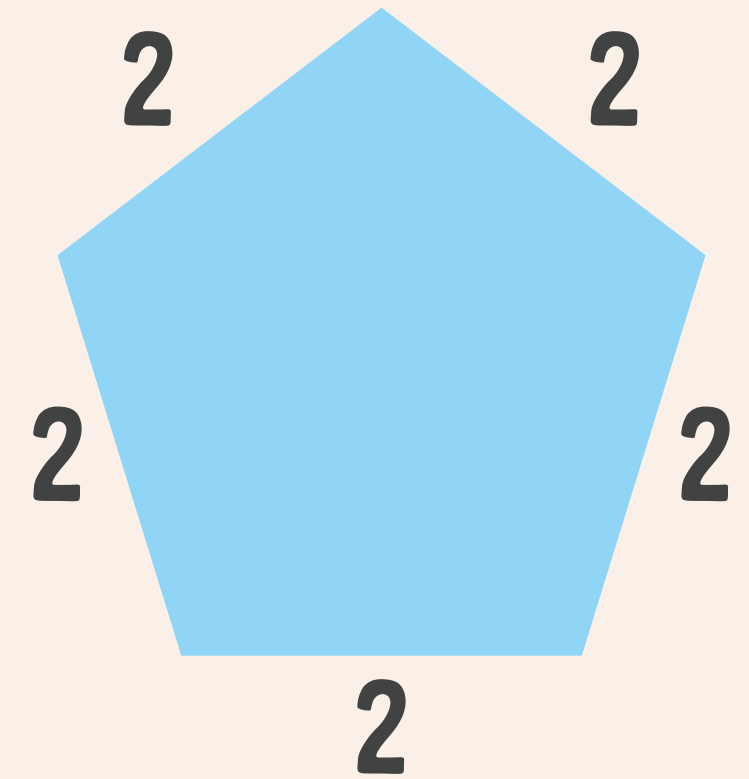
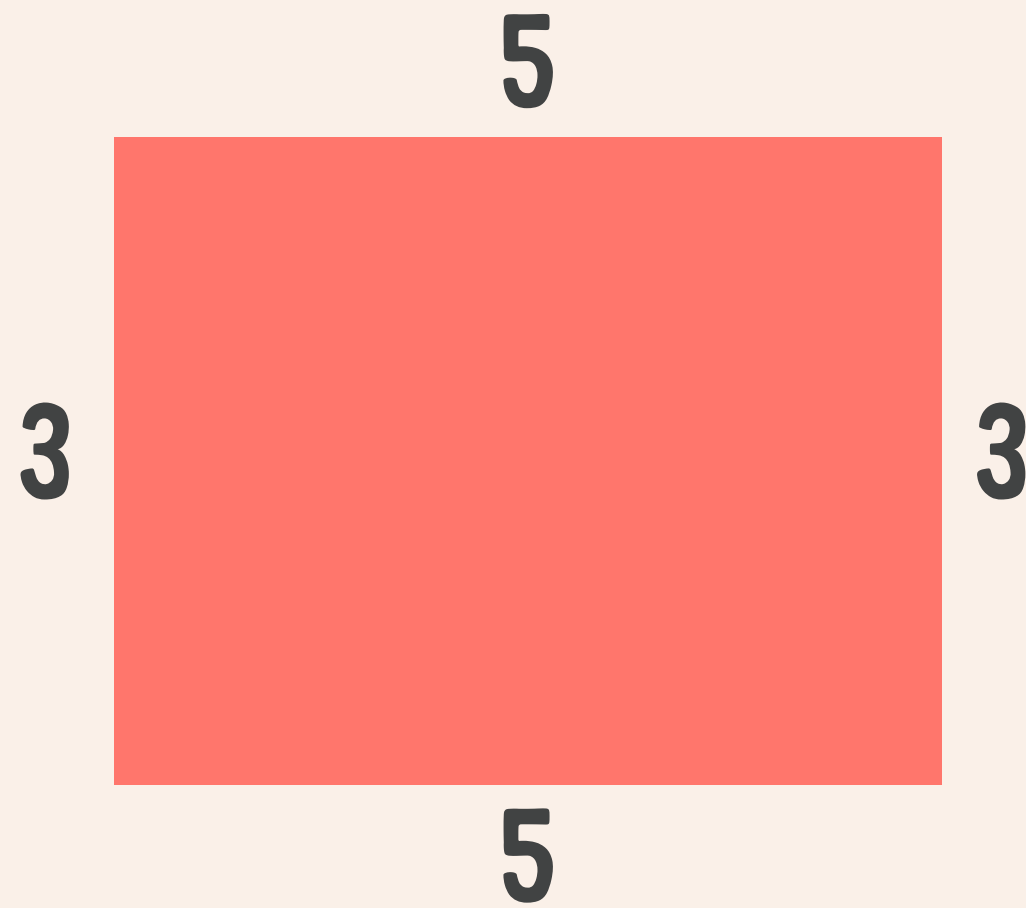
Measure each shape to determine the perimeter and area. Each block represents 1 unit.

Perimeter : 16 units

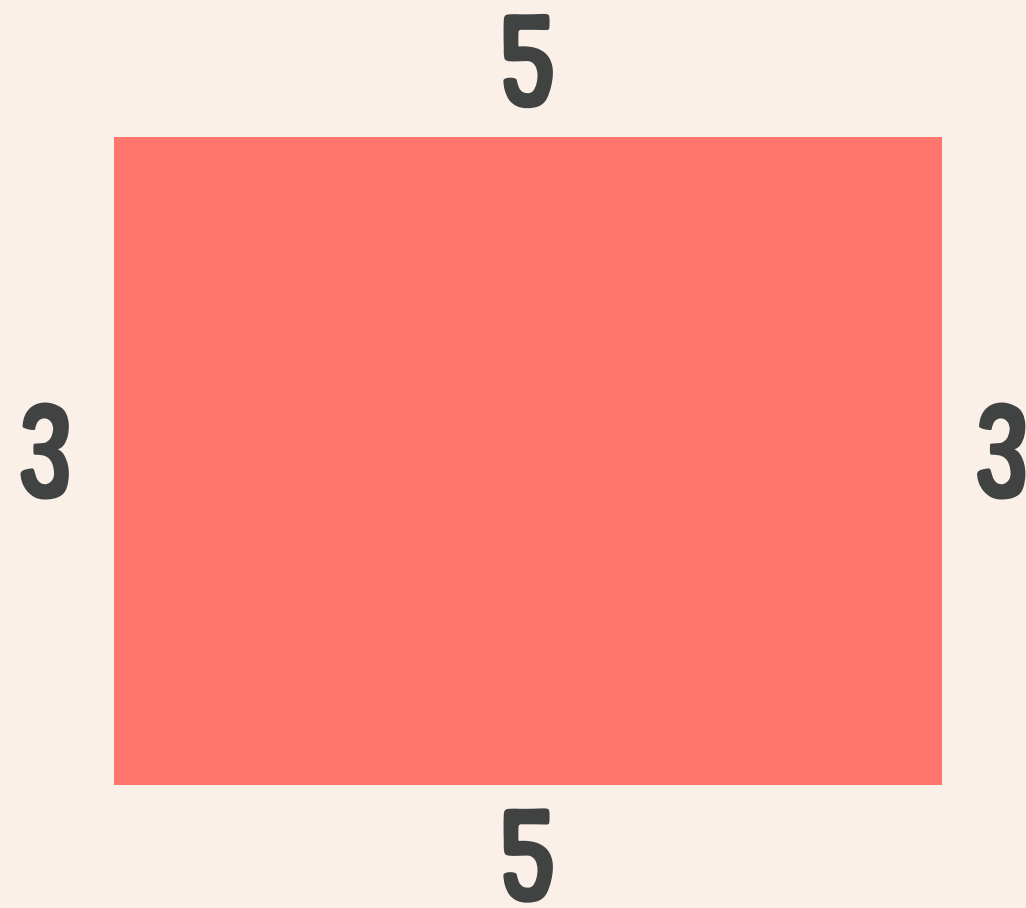
Area: 9 units



What is the Perimeter of These Figures?



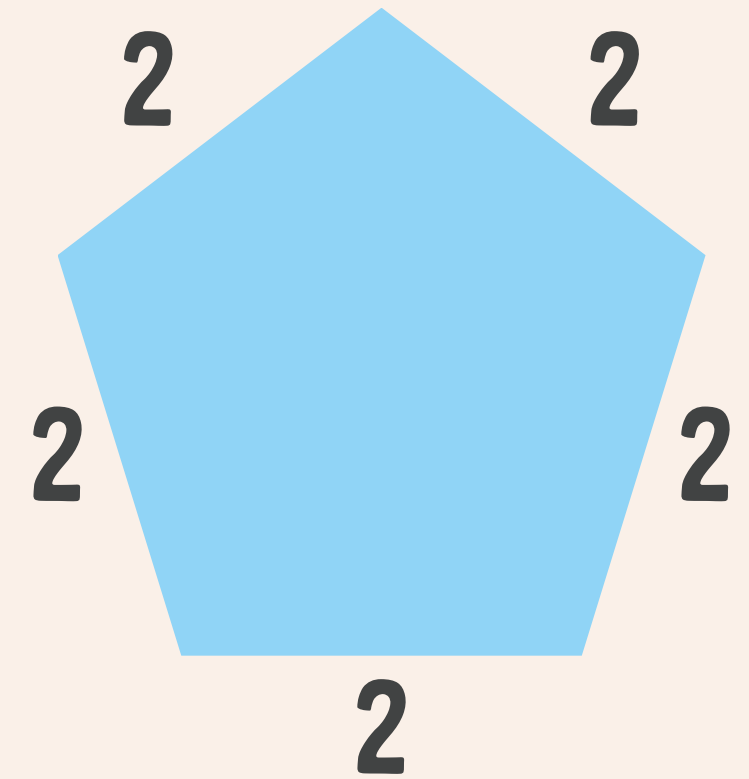
ANSWERS:



$$5 + 5 + 3 + 3 = 16$$



$$1 + 1 + 2 + 2 + 3 = 14$$



$$2 + 2 + 2 + 2 + 2 = 10$$

QUESTION ONE

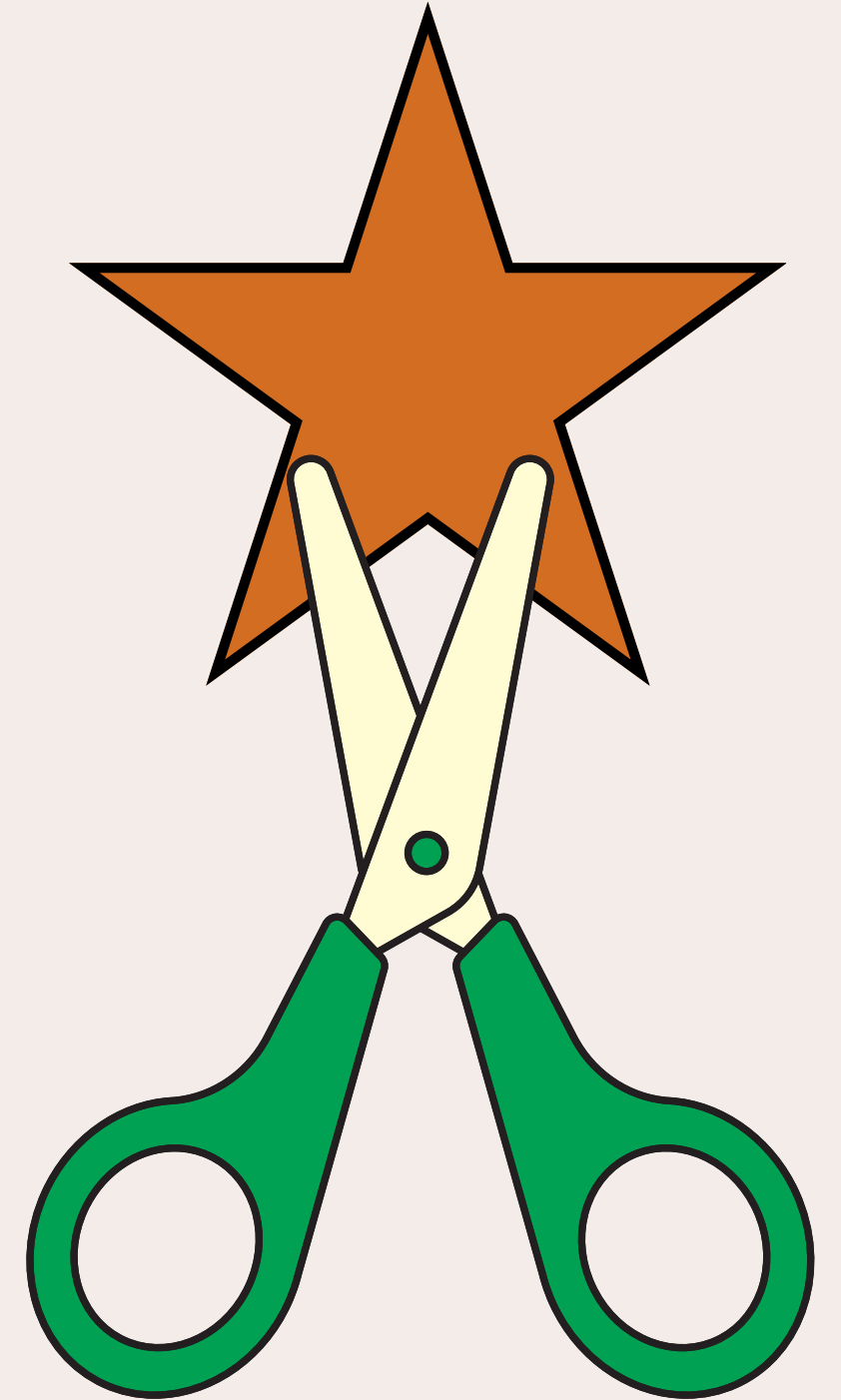
Sarah has a rectangular garden that is 4 meters long and 3 meters wide. What is the perimeter and area of her garden?



QUESTION TWO

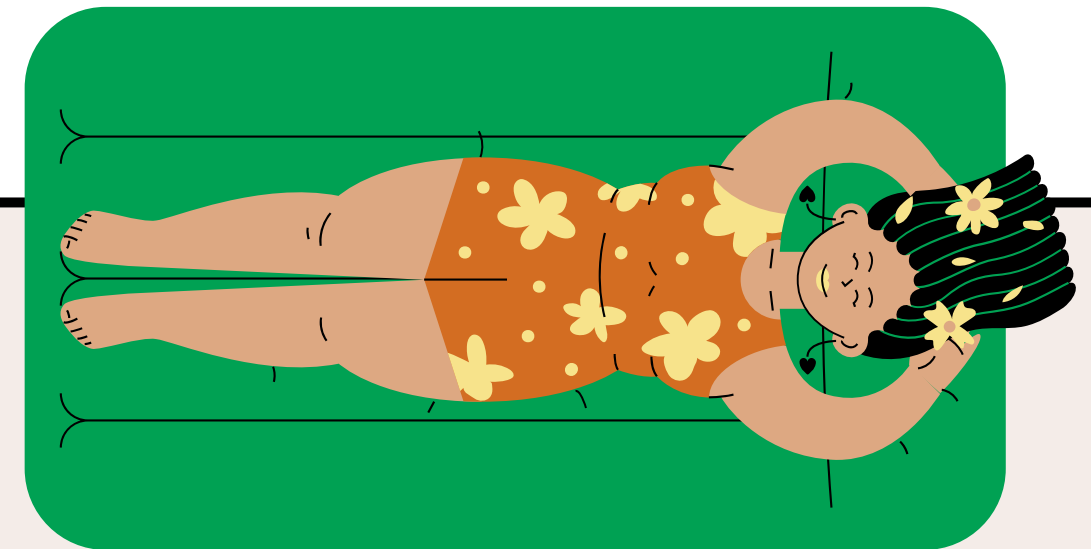
Brendan needs to cut out a 10-sided star at school for his Art History project.

If each side of the star is 4.5 centimetres, what is the perimeter of the star?



QUESTION THREE

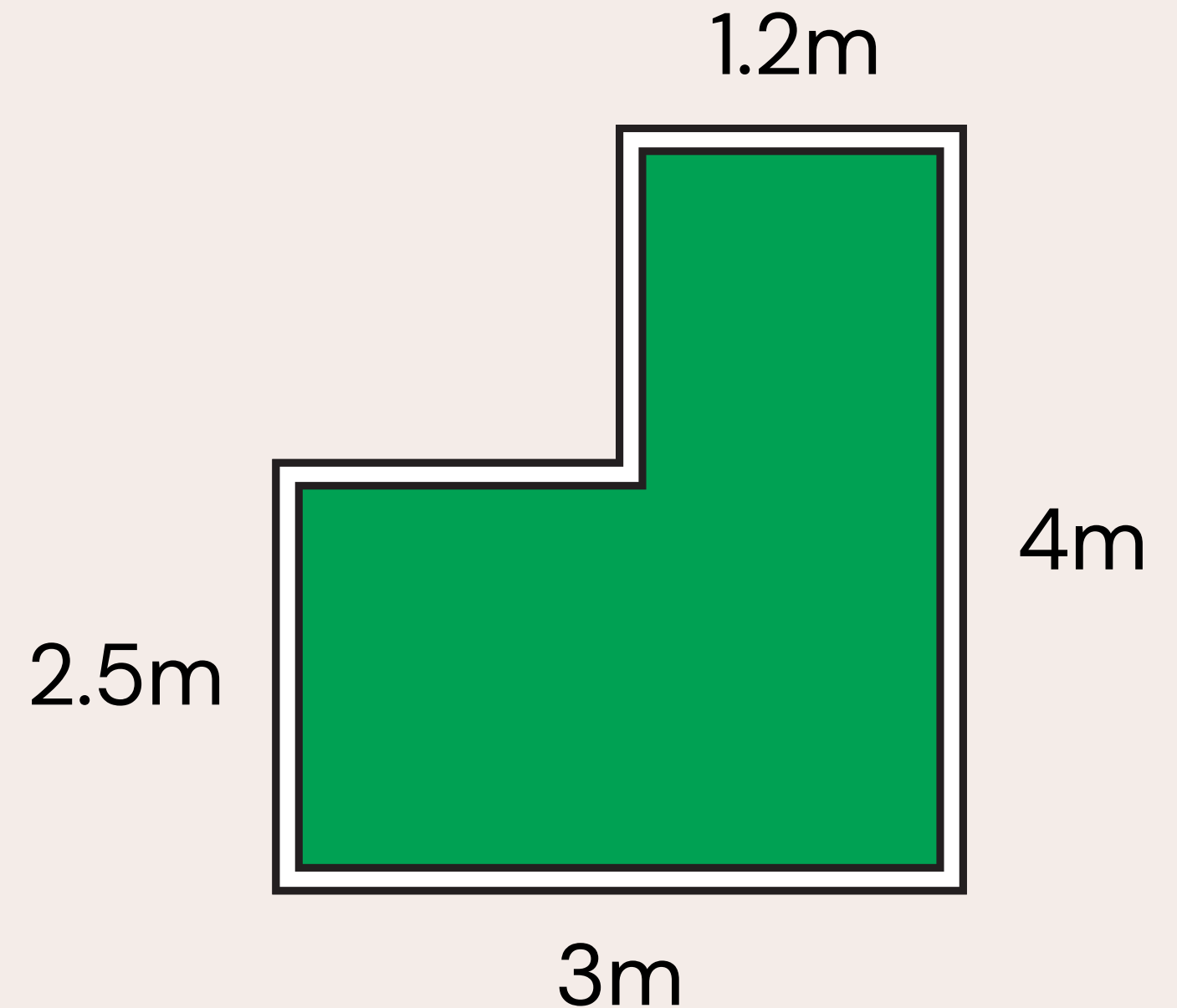
The area of a square swimming pool is 25 square meters
What is the side length of the pool?



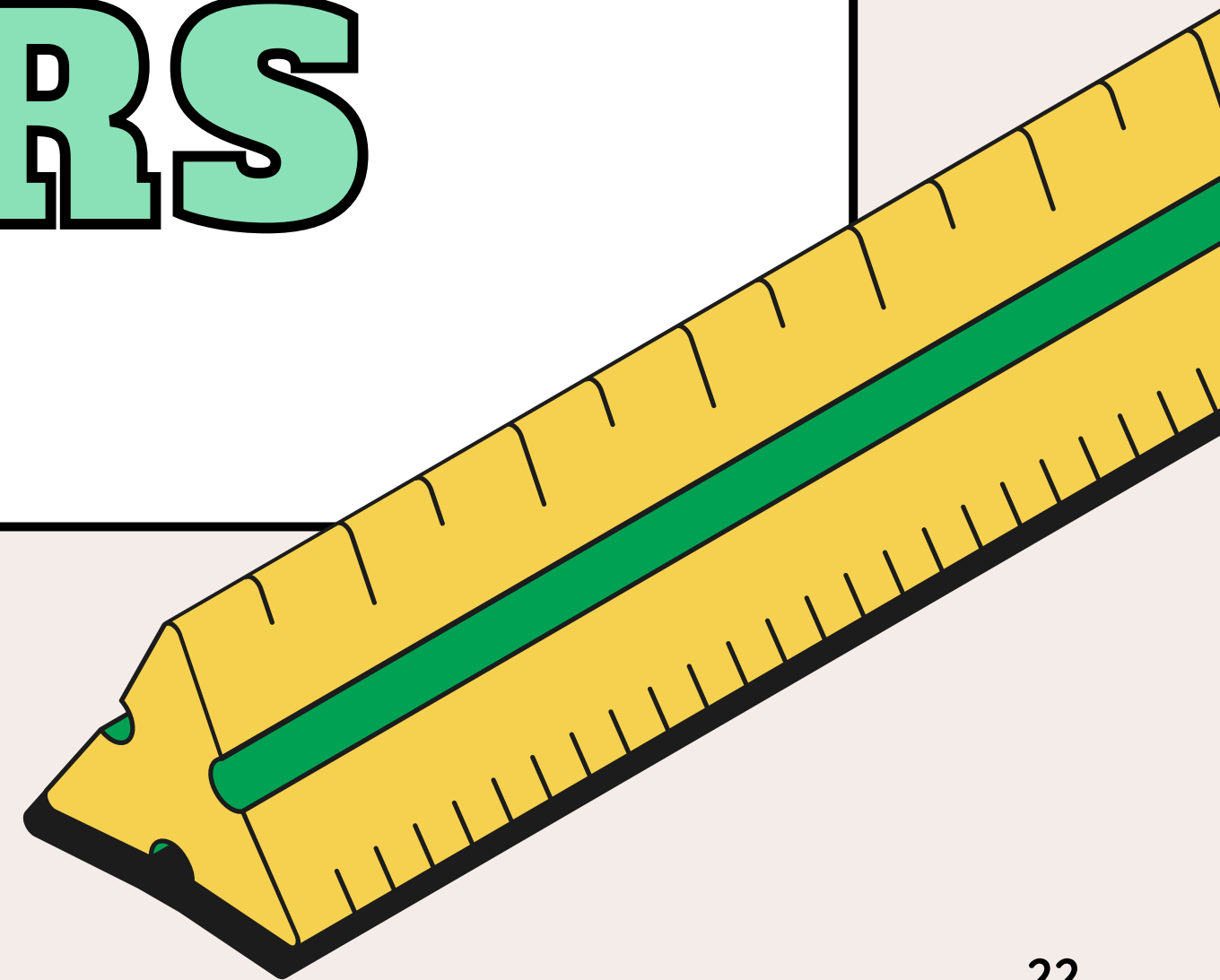
CHALLENGE

QUESTION FOUR

Ciara is fencing around her vegetable garden to keep her cheeky dog out. How much fencing does she need in total?



ANSWERS



QUESTION ONE

Perimeter = the sum of all sides

Perimeter = $4 + 4 + 3 + 3$

Perimeter = 14 metres

Area = length x width

Area = 4×3

Area = 12 square metres



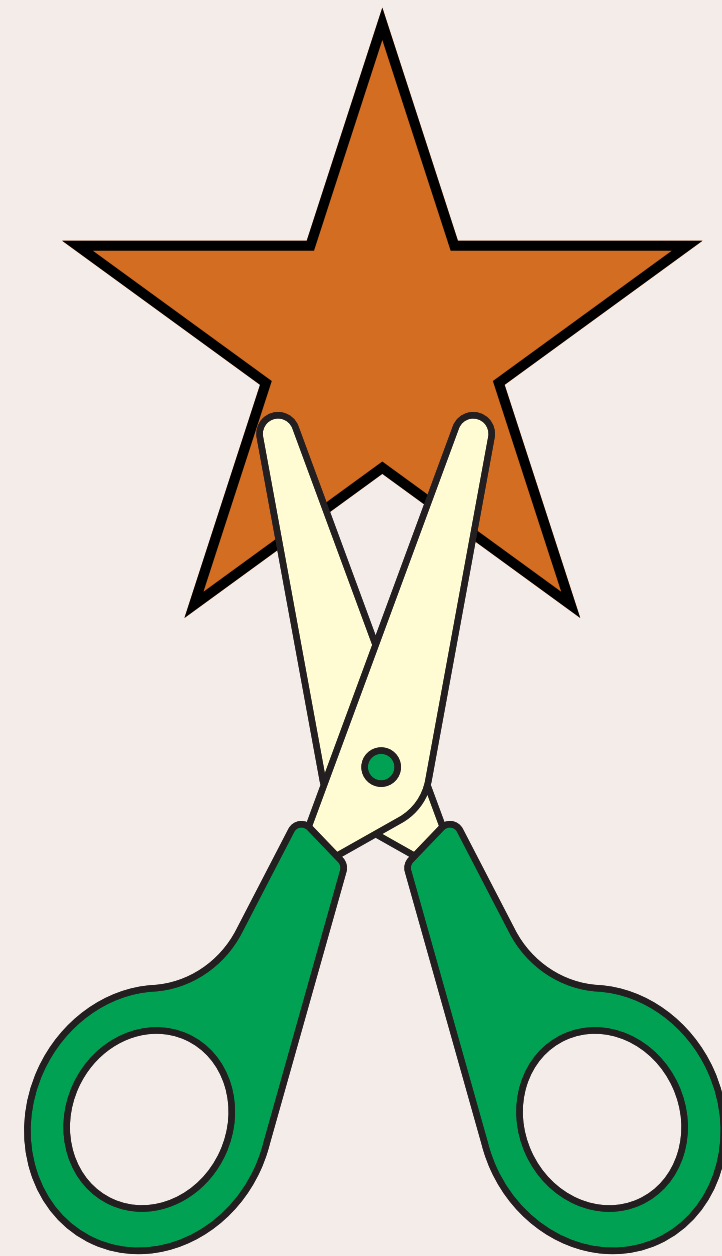
QUESTION TWO

The star has 10 sides. Each side is 4.5cm long.

Perimeter = the sum of all sides

Perimeter = 10×4.5

Perimeter = 45 centimetres



QUESTION THREE

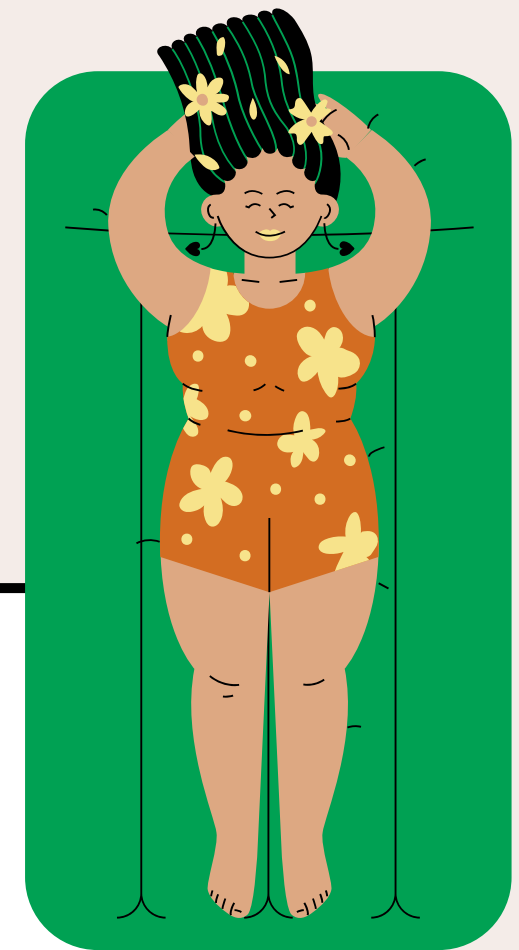
The pool is square.
This means each side is equal.

Area = 25 square metres

Area (square) = l^2

What number multiplied by itself equals 25?

5. This means the pool must be 5 metres by 5 metres.



QUESTION FOUR

Let's find all the side lengths first!

Side A:

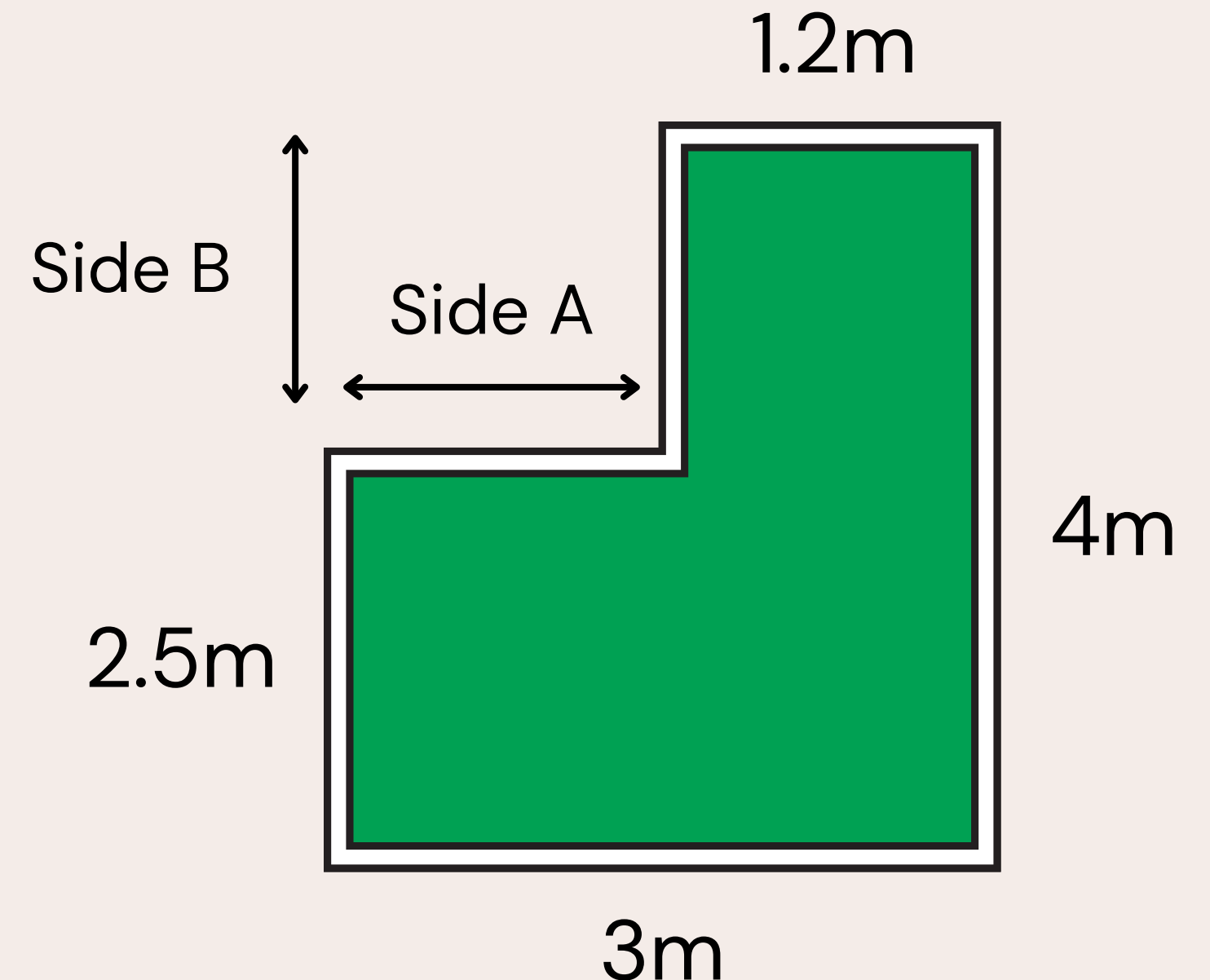
$$\text{Side A} = 3 - 1.2$$

$$\text{Side A} = 1.8\text{m}$$

Side B:

$$\text{Side B} = 4 - 2.5$$

$$\text{Side B} = 1.5\text{m}$$

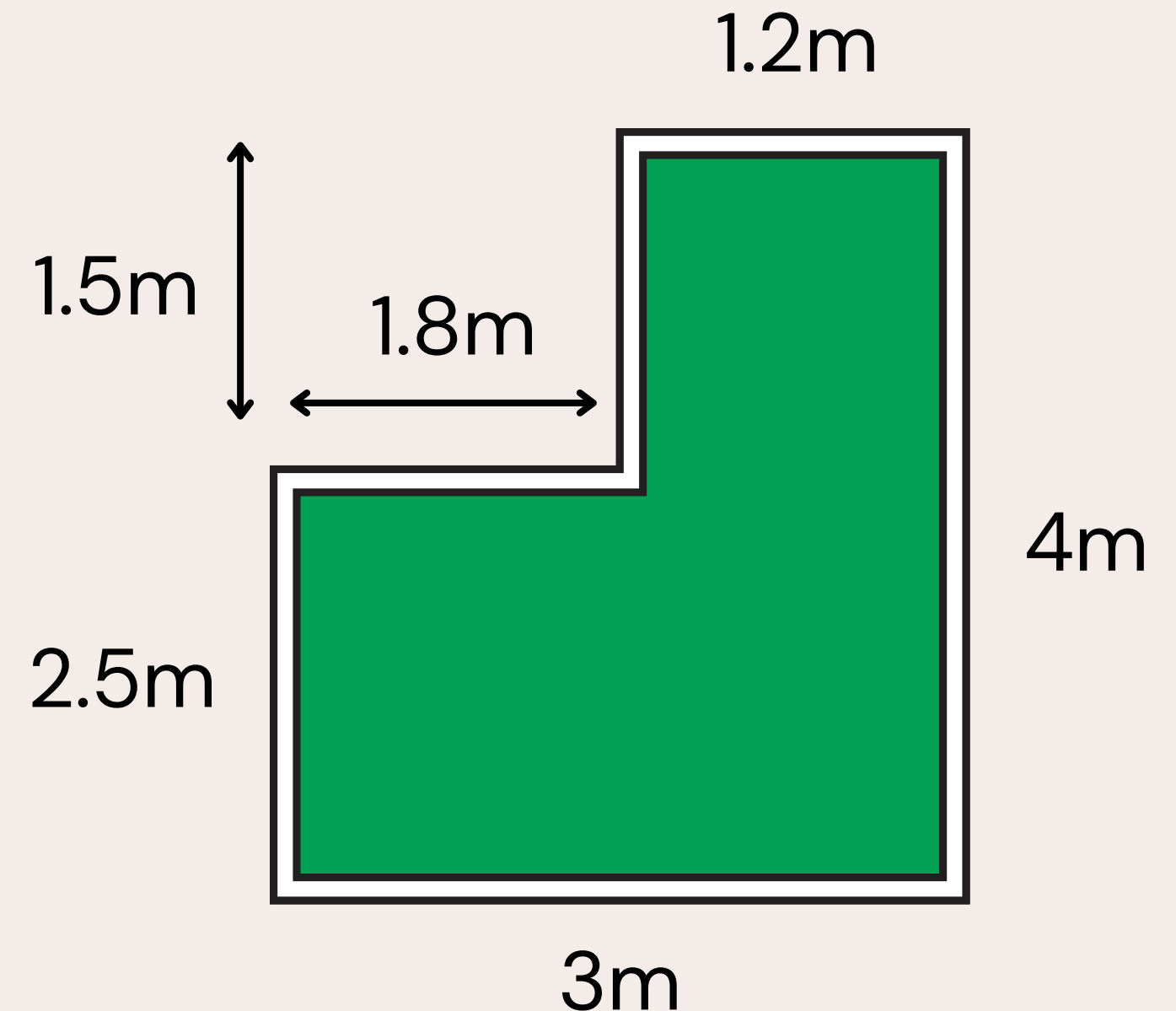


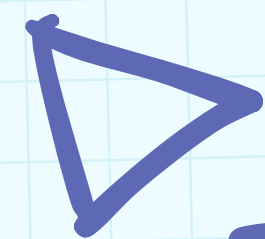
QUESTION FOUR (CONT.)

Perimeter = the sum of all sides

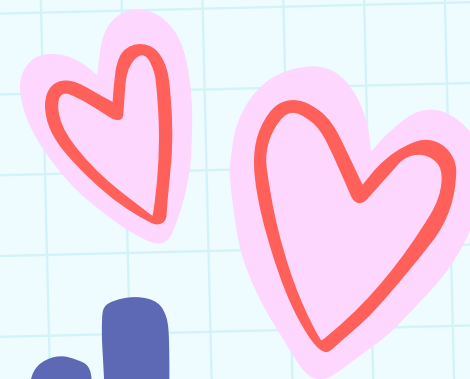
Perimeter = $1.2 + 1.8 + 1.5 + 2.5 + 3 + 4$

Perimeter = 14 metres





Thank You!



Grade 4